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Survival pattern of European hare in a decreasing population

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Summary

1. Age- and sex-dependent survival rates of European hare, *Lepus europaeus*, were estimated for a declining population living in intensive large-scale farming conditions in north-eastern France. During the period 1990–92, a capture–mark–recapture design was used to obtain robust survival rate estimations.

2. The best fit was obtained with survival rates estimated separately for adults and yearlings. Sex dependence was added only in yearlings ($\phi_{\text{male yearling}} = 0{\cdot}471$, $\phi_{\text{female yearling}} = 0{\cdot}237$, $\phi_{\text{adult}} = 0{\cdot}507$ whatever the sex).

3. These estimates and fecundity data obtained from hunting bag analysis and the literature were incorporated into a Leslie Matrix model in order to determine the sensitivity of the population growth rate (λ) to variations of the different demographic parameters.

4. Contrary to that of a population with a higher annual recruitment, the growth rate of the population studied ($\lambda = 0{\cdot}80$), whose annual recruitment was weak, was more sensitive to maintenance than to recruitment variations.

Key-words: *Lepus europaeus*, survival rate, demographic sensitivity, France.

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Introduction

Wildlife management is often based on data that originates from population censuses. Similar population sizes or age-structures may conceal, however, different underlying patterns of population dynamics (Caughley 1974). It is, therefore, desirable to obtain estimates of the basic demographic variables affecting animal numbers, i.e. parameters of maintenance (adult survival) and recruitment (proportion of breeding females, fecundity, juvenile survival). Following this, critical components may be identified to specify population management policies (Emlen & Pikitch 1989). Lagomorphs are usually regarded as reproductive specialists on the so-called *r–K* continuum (Krebs 1986). However, demographic tactics among the different species within this order are not uniform (Flux 1981). Hare and pika life histories can more likely be traced to a *K*-selection than rabbits' (Swihart 1984). Both maintaining and recruitment, therefore, may be critical components of *Lepus* dynamics.

For the European hare *Lepus europaeus* (Pallas), bag records have indicated a marked population decline in several European countries from the 1960s onwards (Pielowski & Pucek 1976; Tapper & Parsons 1984). Numerous factors may reduce hare populations: diseases (Barrat & Biadi 1993) predators (Tapper, Brockless & Potts 1993) and heavy hunting pressure (Pépin 1987) being the ones most commonly put forward. The underlying dynamics seemed highly time and space dependent. None of these factors, therefore, could explain the global negative population trend. As a consequence, an external driving force that would have similar effects on a European scale should be sought. The development of modern agriculture was suggested to be a common denominator (Tapper & Barnes 1986; Slamecka 1991), but few papers give robust estimates of demographic parameters in such environmental conditions, e.g. Hansen (1992) for recruitment components. As far as survival is concerned, only one Danish study provides both robust and age/sex-dependent estimates (Abildgard, Andersen & Barndorff-Nielsen 1972). The ecological conditions, however, differed largely from those herein considered.

A capture–mark–recapture (CMR) experiment was part of a research programme that aimed at studying a possible link between hare decline in France and the decrease of landscape diversity due to farming practices. The CMR design combined with the recent developments of stochastic models, provides more robust estimates of survival rates than those inferred from age-structure analyses (Nichols 1992; Lebreton, Pradel & Clobert 1993). The present work examines survival patterns of hares, by age and sex, in a declining population. Sensitivity of the population growth rate (λ) to changes in demographic parameters was then estimated by resorting to simulations with the Leslie Matrix model.

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Methods

STUDY AREA

The study was carried out on farmlands near Laon (Aisne, north-eastern France). The study area covered 6·5 km^2 in an agricultural region of cereal and sugar beet large-scale farming. The average ground cover was: 45% winter wheat, 30% sugar beets, 9% peas, 8% fallow fields, 6% maize and 2% woodland. Mean field size was 20 ha and linear landscape elements were poorly represented (hedgerows + dirt tracks <50 m ha^{-1}).

DATA COLLECTION

Hares were caught and marked using two small individually numbered metallic ear-tags. All captures took place in December after the hunting season. The first capture operations were carried out in 1990. Recaptures, during which tags were read or placed on animals not previously caught, took place in December 1991 and 1992. All captures and recaptures were performed using exactly the same capture device. Hares were flushed out by a chain of *c.* 250 beaters and captured with 2-km-long nets erected vertically and hung loosely in the fields. Catchers put the hares into cotton bags and a specialized team noted their weight, sex and age. We used the presence/absence of the uniting cartilage of the ulna distal epiphysis to define two age classes: juveniles (<7 months old) and adults (>7 months old). All hares shot on the study area were examined: one eye per animal was removed and kept in a 10% formalin solution. The eye lenses were extracted and weighed after drying for 48 h in an oven at 100 °C. From a standard curve (Broekhuizen & Maaskamp 1979), hares were classified as adults or juveniles for which cohorts were established according to month of birth. The 'ulna palpation' ageing method for living hares may result in some juveniles born early in the season being classified as adults (Walhovd 1966; Suchentruck, Willing & Hartl 1991). In the present case, the analysis of the hunting bag data showed the breeding chronology to be left-censored: no juvenile was estimated to be born before May. The ulna palpation method was thus considered as valid.

POPULATION TREND

Estimates of population numbers in December were obtained using Bailey's modification of the Peterson–Lincoln Index, corrected for annual survival rates (Begon 1979). Hare population numbers amounted to 175 (SD = 27) and 130 (SD = 23) individuals in 1991 and 1992. Because obtaining these estimates required that the animals be previously marked, no figure was available for 1990. An index of abundance (derived from Péroux 1991) showed a marked decline in population size between the beginning and the end of the study: the latter was found to be only one-third of the former. The population under study obviously faced an unbalanced demographic output.

DATA ANALYSIS

Estimating survival rates

Cormack–Jolly–Seber models were used for estimations of survival (ϕ) and recapture (p) probabilities. The CR software (Lebreton *et al.* 1993) proved to be a flexible and powerful tool for modelling survival and testing biological hypotheses: several data sets could be analysed simultaneously allowing the incorporation of individual features, such as age class or sex, and interactions between factors could also be tested using dummy variables. Mathematical backgrounds and detailed examples can be found in Pollock *et al.* (1990) and Lebreton *et al.* (1992). The selection of a model was achieved by comparing the different nested models, using both the Likelihood Ratio Test (LRT) to compare their deviance (DEV), and the Akaike's Information Criterium [AIC = DEV + 2 × (number of parameters)]. When selecting a model with the smallest AIC, both its fit and parsimony were combined (Burnham & Anderson 1992). Estimated annual survival rates of adults were then transformed to mean life span using $0{\cdot}5 + 1/(1 - \phi)$ (Seber 1973). The short duration of the study, 3 years, prevented us from studying changes in ϕ over time. The capture effort was constant over years if estimated as number of beats × number of beaters × length of nets. Since all shot animals were examined and the number of marked ones was known exactly, it was possible to fit models with or without the hunting yield.

Estimating demographic sensitivity

Sensitivity of the population growth rate (λ) to variations of the different demographic parameters was estimated by simulation runs using the Leslie Matrix model (Leslie 1945, 1948; Lebreton 1981). Average values of fecundity and proportion of breeding females were obtained from the literature (Frylestam 1980; Broekhuizen & Maaskamp 1981; Pépin 1989; Hansen 1992). To achieve the age structure observed in the hunting bag, postnatal survival was adjusted iteratively. To compare the population under study with a more current demographic situation, two sets of recruitment parameters were used as starting points. The first one resulted in the observed age structure when estimated from the hunting bag analysis (i.e. 42% of young) whereas the second, hypothetical one, fitted the most often mentioned age structure for the European hare in western Europe (i.e. about 2/3 young) (Broekhuizen 1979; Pépin 1987). When assuming that demographic parameters are constant from one year to another, the population size is multiplied each year by a constant λ. Each parameter was allowed to vary by 5–20% of its initial value and resulting changes in λ were converted to relative sensitivity, i.e $\Delta\lambda/\lambda$. Changes in survival rates due to hunting yield were examined in the same way.

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Results

A total of 258 captures were made during the three sessions, representing 182 individuals: 53 adult males, 65 adult females, 36 juvenile males and 28 juvenile females. The hunting bag for the 3 years amounted to 123 hares: 52 juveniles and 71 adults, 21 of which had been tagged. Survival models were fitted once using the full data set $n = 182$ (with hunting yield), and then using the restricted data set $n = 182 - 21$ (without hunting yield).

EFFECTS OF AGE AND SEX ON RECAPTURE AND SURVIVAL PROBABILITIES

At the time of capture (December), one could assume post-natal and juvenile mortalities to be over. But in the following year, survival of the young might still tend to increase (Abildgard *et al.* 1972). So, all young animals (<7 months old) were considered as yearlings from their release till next capture time (i.e. December year n – December year $n + 1$), and a separate data set from adults was established. Potential sources of variation in survival and recapture rates were therefore age class (yearling, adult), sex, and the interactions between these two factors.

The starting model for yearlings assumed survival (ϕ) to be dependent on age and sex, and capture (p) to be dependent only on sex (ϕ_{a*s}, p_s). For adults parameterization allowed survival and capture rates to be dependent only on sex (ϕ_s, p_s). The initial fully parameterized models fitted the data : χ^2= 1·20 and 0·42, 1 df, with or without hunting yield. Survival rates were the parameters of main interest whereas capture probabilities were considered as noise parameters. The estimation procedure started thus by modelling capture probabilities to increase parsimony and, as a result, the statistical power of further tests on survival rates. Meaning and notation of the various nested models are listed in Table 1.

Annex 1 gives the list of in-between model tests (LRT) and AIC; only models with the smallest AIC are presented hereafter. Indexes to age class and hunting yield (*yg* or *ad* for yearlings or adults, *y* or *ny* for yield or no yield) were used for practical purposes. A first attempt towards parsimony consisted in dropping the effect of sex on p: models with sex-independent capture rates were accepted both for yearlings and adults, with and without hunting yield ($\chi^2_{yg.y} = 0{\cdot}01$, $\chi^2_{ad\,y} = 1{\cdot}33$, $\chi^2_{yg\,ny} = 0{\cdot}01$, $\chi^2_{ad\,ny} = 0{\cdot}74$, 1 df).

For yearlings, parallelism of age effect on ϕ across sexes was accepted whether hunting yield was taken into account or not. Using only the yearling data set, age dependence in ϕ was even non-significant, probably because of small sample size. The hypothesis of sex effect on ϕ was, on the other hand, close to acceptance ($\chi^2_{yg\,y} = 2{\cdot}41$, 1 df, $P = 0{\cdot}12$, $\chi^2_{yg\,ny} = 1{\cdot}69$, 1 df, P = 0·19). As a result, the model including sex dependence in ϕ was retained ($\phi_{male\ yearling} > \phi_{female\ yearling}$, Table 2), because (i) AIC was smallest with ϕ depending on sex, and (ii) in such data modelling the use of a Type I error rate level of $\alpha = 0{\cdot}15$ was recommended as a basis of rejection of a model (Lebreton *et al.* 1992). In this case, the value $\chi^2_{0\,05} = 3{\cdot}84$ is replaced by $\chi^2_{0{\cdot}15} = 2{\cdot}07$.

Modelling adult survival rates resulted in the acceptance of models assuming both ϕ and p to be constant and independent of sex ($\chi^2_{ad\,y} = 0{\cdot}02$, $\chi^2_{ad\,ny} = 0{\cdot}03$, 1 df, NS). When compared under model (ϕ, p), yearling survival rates were on average 25% lower than those of adults ($\phi_{yg.y} = 0{\cdot}392$ $\phi_{yg\,ny} = 0{\cdot}410$ vs. $\phi_{a.y} = 0{\cdot}501$, $\phi_{a\,ny} = 0{\cdot}564$). Such an age-class effect was close to being significant: $P = 0{\cdot}15$ and 0·12 (one-sided tests, with and without hunting yield). Large confidence limits of estimates accounted for this apparent, low, statistical power of tests. However, considering yearlings and adults separately or pooling them gave only a slight difference in AIC (less than one unit). The most meaningful biological model was thus retained by which survival rates were estimated separately for adults and yearlings (Table 2). Corresponding adult mean life span was 2.5 or 2.8 years with, or without, the hunting yield.

SENSITIVITY OF THE POPULATION GROWTH RATE (λ)

With regard to survival, we defined three age classes: juveniles (from birth to December year 1), yearlings (December year 1 to December year 2), and adults. Animals were assumed to breed only from age class 2 onwards. The distribution of eye-lens weights obtained by the

Table 1. Definition of nested models with ϕ = survival probability and p = recapture probability

Model notation	Meaning
ϕ_{a*s}, p_s	Age and sex effect on ϕ, sex effect on p
ϕ_{a*s}, p	Age and sex effect on ϕ, p constant over sex
ϕ_{a+s}, p	No interaction between age and sex effect on ϕ, p constant over sex
ϕ_a, p	Age effect on ϕ, p constant over sex
ϕ_s, p	Sex effect on ϕ, p constant over sex
ϕ, p	ϕ and p constant over age and sex

Table 2. Survival (ϕ) and recapture (p) rates of the European hare with model (ϕ, p) for adults and (ϕ_s, p) for yearlings, 1990–92, northern France (Standart deviation in parentheses)

	Without hunting mortality		With hunting mortality	
Age class	ϕ	p	ϕ	p
Adults				
M&F	0·564	0·273	0·507	0·339
	(0·191)	(0·127)	(0·151)	(0·135)
Yearlings				
M	0·484	0·470	0·471	0·482
	(0·179)	(0·224)	(0·170)	(0·222)
F	0·271	0·470	0·237	0·482
	(0·149)	(0·224)	(0·135)	(0·222)

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hunting bag analysis gave an age structure of 42% juveniles and 58% other classes. Two hare populations with differing age structures were compared: the observed one (42% juveniles) and a hypothetical but more typical one (67% juveniles) thereafter denoted *OP* and *HP*. The initial sets of recruitment parameters were chosen to obtain these structures when combined with the estimated survival rates (Table 3).

The above parameters led to values of $\lambda = 0{\cdot}80$ and $1{\cdot}30$ for *OP* and *HP*. Each year, *OP* size decreased by 20% whereas *HP* size increased by 30%. If any of the recruitment variables changes by a given percentage for all age classes, this would lead to exactly the same percentage change in λ. Pooling of all these variables enabled us to define the internal recruitment of the population (i.e. proportion of breeding females + fecundity + juvenile survival). Yearling and adult survival rates made up the maintaining component of the dynamics. A 10% decrease in the recruitment led to a 3·6% decrease of λ in the *OP* case and 6·0% for the *HP* case. On the other hand, a decrease of 10% in maintenance resulted in a 6·4% decrease of λ in the *OP* case and 4·0% for the *HP* case. The studied population was more sensitive to fluctuations in maintenance than recruitment parameters (Fig. 1, solid symbols) contrary to the hypothetical population (Fig. 1, open symbols). The differential sensitivity of λ increased with larger changes in demographic parameters but was less important for the hypothetical population than for the observed one. Estimates of ϕ without hunting yield gave a 5·9% increase in λ for the *OP* case as opposed to 2·8% for the *HP* case.

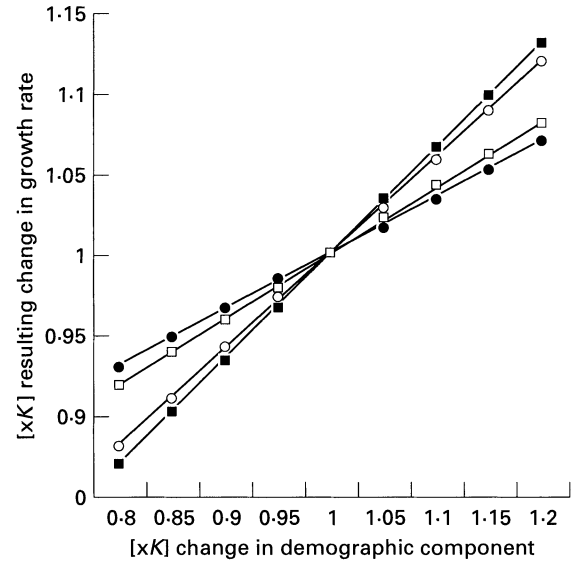

Fig. 1. Sensitivity of the growth rate (λ) to changes in recruitment (• or o) and maintenance (□ or ■) components according to age structure. Two hare populations are compared: the population under study, made of 42% juveniles and 58% adults (solid symbols) vs. a 'classic' hare population, i.e. made of 2/3 juveniles and 1/3 adults (empty symbols). In the former, sensitivity of λ is larger for changes in the maintenance component than in the recruitment one, contrary to the latter.

Discussion

The average annual survival rate in adult hares as obtained from our capture–mark–recapture experiment was $\phi = 0{\cdot}51$ (including hunting yield). Survival of yearlings was lower than that of adults and seemed to be sex-dependent: male yearlings survived better than female yearlings did (0·47 as opposed to 0·24, using a type I error $\alpha = 0{\cdot}15$, see above). Abildgard *et al.* (1972), in Denmark, obtained quite robust estimates of sex-dependent survival rates using a mark–recapture design and Darroch's probabilistic model (ref. in Abildgard *et al.* 1972). In yearlings they also noted a marked tendency towards greater mortality in females than in males ($\phi_{\text{male yearling}} = 0{\cdot}56$, $\phi_{\text{female yearling}} = 0{\cdot}44$).

Table 3. Initial values of demographic parameters used to study the population growth rate sensitivity

Class of first reproduction	2		
Age classes	1 (Juveniles)	2 (Yearlings)	3 (Adults)
Breeding females (%)	–	0·85	0·90
Fecundity	–	10·0	12·5
Survival rates	0·067(*) 0·183(†)	0·392	0·507
Age structure (%)	42·3 66·7	57·9* 33·3†	

* Studied hare population.
† Typical hare population.

Differential trade-offs for energy allocation among sex classes very often occur in the wild (Stearns 1992) and may result in sex-specific survival patterns. European hare does can reach puberty at an early age (Caillol *et al.* 1992) and produce very well developed neonates, whose high postnatal growth rate permits a short nursing period and a high weight at weaning (Broekhuizen & Maaskamp 1980). So both prenatal and neonatal maternal investment are high (Swihart 1984; Pontier, Gaillard & Allainé 1993), particularly for primiparous does. A lower survival for female yearlings than for male yearlings could thus be logically expected. One could also think that the lower ϕ for young females could arise from sex-biased dispersal in their favour. Dispersal and mortality could not be separated in the CMR experimental design we used. However, when a sex-biased dispersal occurs in small mammal species with a promiscuous mating system, such as the European hare, it is more often male-biased (Pusey 1987; Wolff 1994). In our study, out of 45 hares radio-tagged for complementary purposes, only two male yearlings left the study area. Douglas (1970) found similar results using recaptures of ear-tagged hares in New Zealand. Finally, Abildgard *et al.*'s(1972) study took place on Illumø Island where no dispersal was possible. The lower ϕ for female yearlings herein estimated may, therefore, be considered as actual.

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Survival of the European hare has been estimated in several countries. The values obtained, and the resulting mean life span, are variable (Table 4). Part of this variation arises from hunting schemes that differed from one study to another. So did the robustness of the estimates, several of them stemming from the use of horizontal life tables whereas others were simply derived from an analysis of the young *per* adult ratio in the bag. Estimates obtained with both methods do not take into account age-specific mortality and fecundity rates. The second method assumes unchanging dynamics and age structure. The present estimates are based on recent advances in capture modelling, which allow us (i) to separate noise parameters (probability of capture) from interesting ones (survival rates), and (ii) to fit biologically meaningful models (Lebreton *et al.* 1992). Although the population under study was decreasing, adult survival rates and mean life span here estimated were higher than several estimated values when compared to European data. We might hastily conclude that the maintenance component was not responsible for the observed population decrease.

Sensitivity of the population growth rate ($\Delta\lambda/\lambda$) was conditional upon the age structure. Depending on whether it favours adults or juveniles, the changes in λ were more sensitive to variations in parameters of either maintenance or recruitment — the generation time T ranging from 1·6 to 2·8 years. Populations of short-lived species (T-values ≤ 2) display a strong sensitivity of λ to parameters of recruitment, contrary to those of long-lived ones (T-values ≥ 2) for which λ is more sensitive to parameters of maintenance (Lebreton & Clobert 1990). The European hare species could belong to both categories, its position along the continuum depending on the age structure of the population. A larger variability is observed, however, for young production ($CV = 30\%$, from Table 5 in Hansen 1992) than for adult survival ($CV = 17\%$, from Table 4, this study). Therefore, the differential sensitivity of λ may be counterbalanced by a variability that differs between the recruitment and the maintaining components.

The age-ratio recorded — less than one young per adult — indicated that recruitment was not successful in this population. In the European hare, reproductive success is highly variable in time and space (Pielowski 1976; Broekhuizen 1979), depending on the proportion of females producing 0, 1, 2, ..., 6 litters (Hansen 1992) and on postnatal mortality (Pépin, 1989). The age ratio should be considered with caution when used alone because it is only a global measure of breeding success, more or less biased by adult survival (Caughley 1974). In the present study, estimates of population numbers showed a declining trend although adult survival was good. Young specimens made up less than half of the bag and the breeding chronology estimated from the distribution of eye-lens weights, was left-censored: all young specimens in the bag were born between May and October with a peak during August (Fig. 2), whereas hares usually breed from January till October (Hewson & Taylor 1975; Frylestam 1980; Broekhuizen & Maaskamp 1981). Without an analysis of placental scars, one cannot distinguish between the two following scenarios: (i) dispersal or death of young hares born early in the season, or (ii) late breeding.

In any case, the population decline very probably results from this weak recruitment. So diseases, predation or unfavourable ecological conditions (like those resulting from modern agriculture), could still account for root causes, only if regarded as factors lowering recruitment. On the other hand, the resulting age structure, with a deficit of young specimens, made the population very sensitive to later changes in adult survival rates (as for the rabbit species, Smith & Trout 1994), such as those related to hunting.

Some hare populations withstand substantial annual shooting culls (Stoate & Tapper 1993), so several authors have suggested the impact of hunting mortality to be weak. They argue that compensatory mechanisms such as the density-dependent regulation of recruitment could counterbalance hunting exploitation (Broekhuizen 1979; Frylestam 1980). Had our hare population not been submitted to hunting mortality (i.e. ϕ_{adult} increasing from 0·51 to 0·56), λ would have increased by 5·9% (notice that in the present case hunting after-effects mortality is not included). Had a similar 10% increase been assigned to one of the recruitment components (i.e. juvenile survival or fecundity), the impact would have been half as much. Such a differential sensitivity, which was due to the weak annual recruitment of the studied population, should have forced compensatory regulation — should it exist — to be very efficient. However, had hunting mortality been cancelled, the population would have kept on decreasing ($\lambda_{no\ hunting} = \lambda_{hunting} \times 1{\cdot}059 = 0{\cdot}85$, still lower than unit), so hunting can be regarded mainly as an aggravating factor.

The apparent lack of young born early in the breeding season and the consecutive peak of births of surviving hares in summer are striking and would certainly constitute the more pertinent subject for further research on this population problem. Source–sink systems (Pulliam 1988) may also dampen or amplify changes in population numbers. The next step in managing species as harvestable resources, therefore, will probably require the use of subdivided population concepts and models (e.g. Lebreton & Gonzalez-Davila 1993). Whatever the theoretical approach used, the fact remains that once a hare population has poor

Table 4. Survival rates (ϕ) and mean life span ($0.5 + 1/1 - \phi$) in years of adult brown hares from literature (Abildgard *et al.* 1972; Broekuizen 1979; Kovacs 1983; Pépin 1987)

Country	ϕ	Mean life span	Hunting
Denmark	0·59	2·92	No
Poland	0·51	2·50	No
France	(0·56)*	2·75	No
	0·51	2·50	Yes
Poland	0·52	2·58	Yes
Netherlands	0·35	2·00	Yes
Hungary	0·38	2·08	Yes
France	0·40	2·17	Yes

* Present study when excluding hunting mortality.

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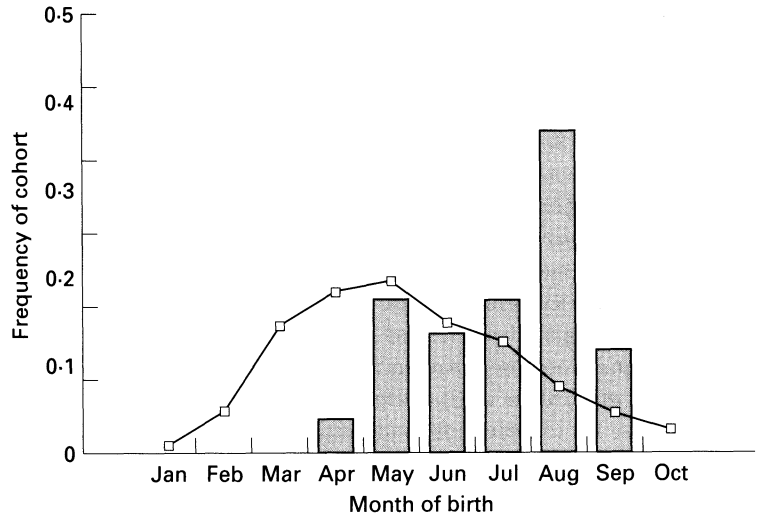

Fig. 2. Monthly cohort of birth of shot juvenile hares as estimated from eye lens weights (J, F, M...: January, February, March...). The distribution is left-censored in the studied population (bars) as opposed to a more typical one (□ and full line; from Hewson & Taylor 1975).

recruitment, its growth rate becomes very sensitive to any kind of additional mortality imposed on the breeding adults, such as that related to hunting.

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Annexe 1

Results of capture–recapture models and comparisons between them. For each one were given the number of parameters (np), the deviance (DEV) and the Akaike Information Criterion (AIC)

Model		np	DEV	AIC	Comparison
I. Including hunting mortality					
(A) Basic models					
(i) Yearlings	(1) (ϕ_{a*s}, p_s)	6	79·83	91·84	ϕ and p depending on sex both for yearlings and adults. Age dependence
(ii) Adults	(2) (ϕ_s, p_s)	4	157·84	165·84	added on ϕ for yearlings
(B) Modelling capture rate					
					Sex effect on p
(i) Yearlings	(3) (ϕ_{a*s}, p)	5	79·84	89·84	(3) vs. (1) $\chi^2_1 = 0{\cdot}01$, $P = 0{\cdot}920$
(ii) Adults	(4) (ϕ_s, p)	3	159·17	165·17	(4) vs. (2) $\chi^2_1 = 1{\cdot}33$, $P = 0{\cdot}249$
(C) Modelling survival rate					
					Parallelism of age across sex
(i) Yearlings	(5) (ϕ_{a+s}, p)	4	81·75	89·75	(5) vs. (3) $\chi^2_1 = 1{\cdot}91$, $P = 0{\cdot}167$
	(6) (ϕ_a, p)	3	83·90	89·90	Sex effect on ϕ
					(6) vs. (5) $\chi^2_1 = 2{\cdot}15$, $P = 0{\cdot}143$
	(7) (ϕ_s, p)	3	81·77	87·77	(8) vs. (7) $\chi^2_1 = 2{\cdot}41$, $P = 0{\cdot}121$
					Age effect on ϕ
	(8) (ϕ, p)	2	84·18	88·18	(7) vs. (5) $\chi^2_1 = 0{\cdot}02$, $P = 0{\cdot}888$
					(8) vs. (6) $\chi^2_1 = 0{\cdot}28$, $P = 0{\cdot}597$
					Sex effect on ϕ
(ii) Adults	(9) (ϕ, p)	2	159·19	163·19	(9) vs. (4) $\chi^2_1 = 0{\cdot}02$, $P = 0{\cdot}887$

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Annexe 1 (*Continued*)

Model		np	DEV	AIC	Comparison
II. Without hunting mortality					
(A) Basic models					
(i) Yearlings	(1) (ϕ_{a*s}, p_s)	6	73·41	85·41	ϕ and p depending on sex both for yearlings and adults. Age dependence
(ii) Adults	(2) (ϕ_s, p_s)	4	135·26	143·26	added on ϕ for yearlings
(B) Modelling capture rate					
					Sex effect on ϕ
(i) Yearlings	(3) (ϕ_{a*s}, p)	5	73·43	83·42	(3) vs. (1) $\chi^2_1 = 0{\cdot}01$, $P = 0{\cdot}920$
(ii) Adults	(4) (ϕ_s, p)	3	136·00	142·00	(4) vs. (2) $\chi^2_1 = 0{\cdot}74$, $P = 0{\cdot}390$
(C) Modelling survival rate					
					Parallelism of age across sex
(i) Yearlings	(5) (ϕ_{a+s}, p)	4	75·49	83·49	(5) vs. (3) $\chi^2_1 = 2{\cdot}07$, $P = 0{\cdot}150$
	(6) (ϕ_a, p)	3	76·74	82·74	Sex effect on ϕ
					(6) vs. (5) $\chi^2_1 = 1{\cdot}25$, $P = 0{\cdot}264$
	(7) (ϕ_s, p)	3	75·85	81·85	(8) vs. (7) $\chi^2_1 = 1{\cdot}69$, $P = 0{\cdot}194$
					Age effect on ϕ
	(8) (ϕ, p)	2	77·54	81·54	(7) vs. (5) $\chi^2_1 = 0{\cdot}36$, $P = 0{\cdot}549$
					(8) vs. (6) $\chi^2_1 = 0{\cdot}80$, $P = 0{\cdot}371$
					Sex effect on ϕ
(ii) Adults	(9) (ϕ, p)	2	136·03	140·03	(9) vs. (4) $\chi^2_1 = 0{\cdot}03$, $P = 0{\cdot}862$

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